

Decision Trees and Pegasos SVMs from Scratch

Price and price-tier prediction on bina.az listings

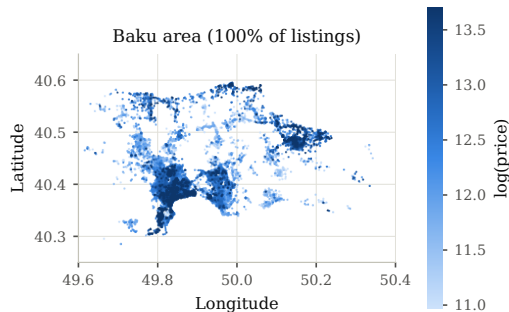
Surname1, Name1 Surname2, Name2 Surname3, Name3

MLE-AI-201 Machine Learning — AI Academy

Code: github.com/eyyubovsamur240-afk/ML-final-project (tag v1.0-final)

1. Problem and dataset

- bina.az sale listings (Kaggle): 100,775 raw rows $\rightarrow$ 57,721 clean
- **Task A:** regression of $\log(\text{price})$ (skewness 6.60 $\rightarrow$ 0.32 after the log)
trees learn log price per m^2 , then add log area back
- **Task B:** premium vs. standard, split at the *training* median (208,000 AZN)
- 70/15/15 split, stratified on price deciles: 40,405 / 8,658 / 8,658
- Tree and SVM are **our own NumPy code**; scikit-learn only as a baseline



2. Cleaning, features and leakage

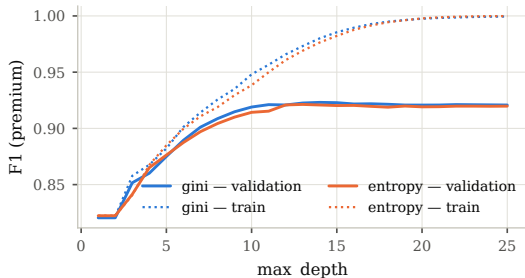
- Azerbaijani column names → English; strings like “85 m²”, “5 / 9”, “1 250 000” parsed
- Repeated scrapes: 100,775 rows but 64,454 listings (latest kept); residential only
- Duplicates and re-posts removed; fixed plausibility rules
- **Leakage dropped:** unit_price, total_price (both functions of price), plus identifiers and promo flags
- Imputation, one-hot levels, scaling and the tier threshold are fitted on **train only**
- 292 features: size, floors, coordinates, distance to centre, flags, district one-hot, metro/settlement tags

Step	Rows left	Removed
raw rows	100,775	
repeat scrapes of a listing removed (latest kept)	64,454	36,321
residential categories only	58,452	6,002
has a positive price	58,452	0
has a positive area	58,452	0
exact duplicates removed	58,110	342
re-posted listings removed	57,737	373
price in [5,000, 15,000,000] AZN	57,728	9
area in [10, 3000] m2	57,724	4
price/m2 in [30, 25,000] (typo filter only)	57,721	3
final cleaned rows	57,721	0

3. Our decision tree

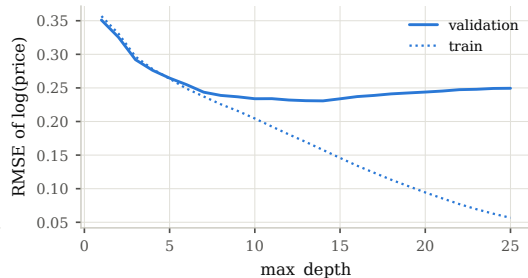
- Gini / entropy / MSE; gain
 $= I(p) - \sum_c \frac{N_c}{N} I(c)$
- Exact, vectorised search: one sort per node, every threshold scored with cumulative sums

(a) Task B: Gini vs entropy



- max_depth, min_samples_split/leaf, min_impurity_decrease
- Chosen on validation: A: depth 14, leaf 5; B: gini, depth 14 (Gini vs. entropy $\Delta F1 = 0.002$)

(b) Task A: MSE criterion

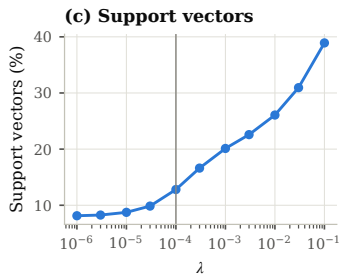
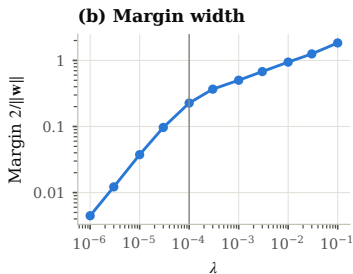
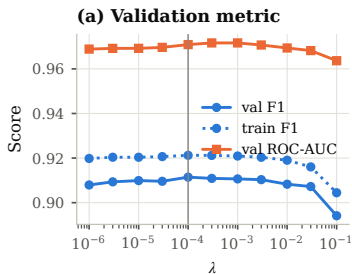


4. Our Pegasos SVM: regularisation and margin

$$J(w) = \frac{\lambda}{2} \|w\|^2 + \frac{1}{n} \sum_i \max(0, 1 - y_i w^\top x_i)$$

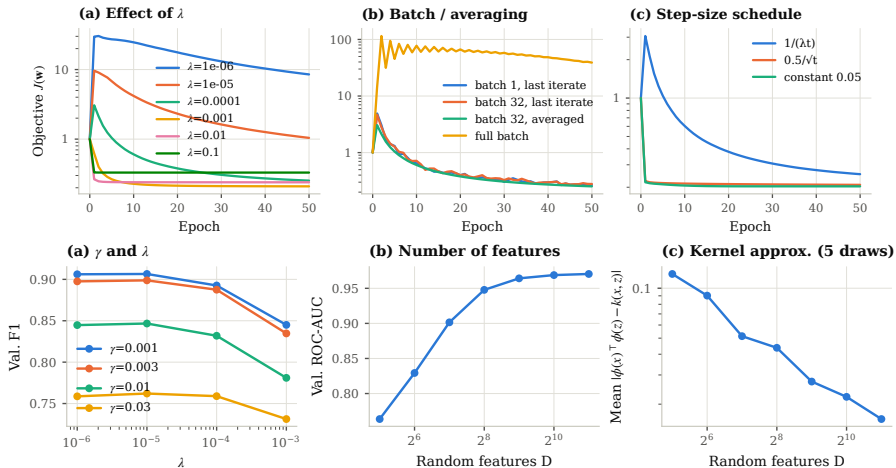
$$w \leftarrow (1 - \eta_t \lambda) w + \frac{\eta_t}{k} \sum_{i \in A_t^+} y_i x_i, \quad \eta_t = \frac{1}{\lambda t}$$

- Mini-batch 32, projection, t -weighted averaging
- $\lambda = 10^{-4}$: margin $2/\|w\| = 0.429$, 19.3% support vectors



4b. Convergence and the RBF extension

- J : 1.000 $\rightarrow$ 0.2053, vs. LIBLINEAR's exact optimum 0.2045 (gap 0.40%)
- Random Fourier features: $\phi(x) = \sqrt{2/D} \cos(Wx + c)$, then the same linear solver ($\gamma = 0.001$, $D = 1,024$)



5. Experimental setup

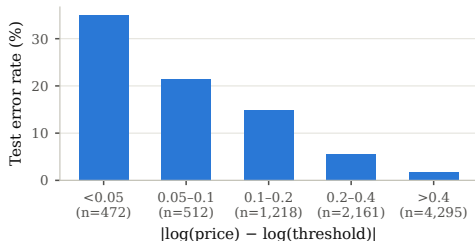
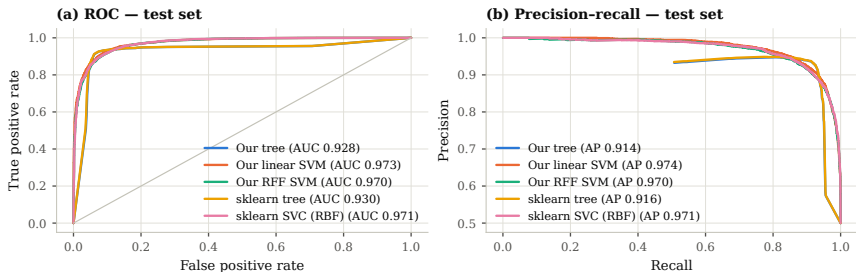
- Every hyperparameter selected on **validation**; refit on train+val; **test scored once**
- Uncertainty: 5-fold stratified CV (preprocessing refitted per fold) + bootstrap CIs + paired bootstrap tests between models
- Task A: RMSE / MAE / R^2 on log price (+ AZN); Task B: P / R / F1, ROC-AUC, PR-AUC
- Matched baselines: `DecisionTree*`, `SGDClassifier(hinge)`, `LinearSVC`, `SVC(RBF)`, with scaling inside a `Pipeline`
- Seed 42 everywhere; `python -m src.run_all` regenerates every number

6. Results

Model	A: RMSE (CV)	A: test	B: F1 (CV)	B: test
Ours: decision tree	0.232±0.003	0.224	0.923±0.003	0.925
Ours: linear SVM (Pegasos)	n/a	n/a	0.917±0.003	0.915
Ours: RBF SVM (RFF + Pegasos)	n/a	n/a	0.913±0.003	0.910
Ours: ridge (bonus)	0.239±0.002	0.231	n/a	n/a
sklearn DecisionTree*	0.232±0.003	0.224	0.922±0.004	0.925
sklearn LinearSVC	n/a	n/a	0.917±0.003	0.915
sklearn SVC (RBF)	n/a	n/a	0.915±0.004	0.911

- Our tree vs. sklearn: 0 gain mismatches, 98.1% test agreement (differences: exact gain ties and scikit-learn's float32 cast of the features)
- Averaged Pegasos F1 0.915 vs LIBLINEAR 0.915 (same objective); RFF falls short of the linear SVM (−0.006 F1)
- Tree wins on F1 but its near-pure leaves give poor ranking: AUC 0.928 vs SVM 0.973
- Paired bootstrap on the test rows: 2 of 12 ours-vs-sklearn gaps are significant; tree — sklearn RMSE −0.0007 [−0.0021, +0.0002]

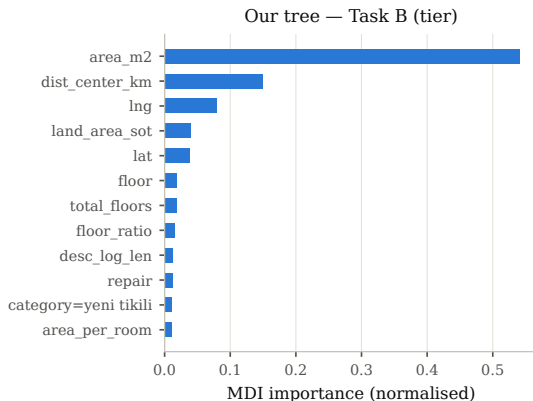
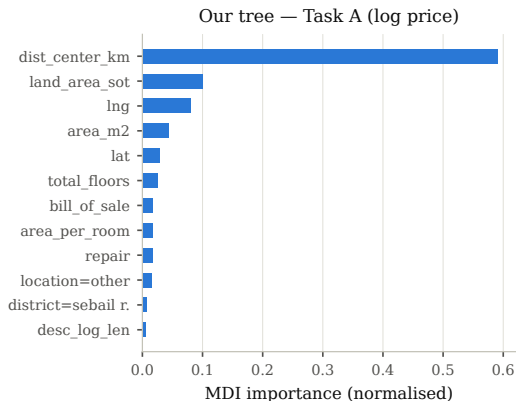
6b. One figure: the classifier is unsure only near the threshold



42% of the tree's errors are listings within $\pm 10\%$ of the threshold price, a band that holds only 11% of the test listings.

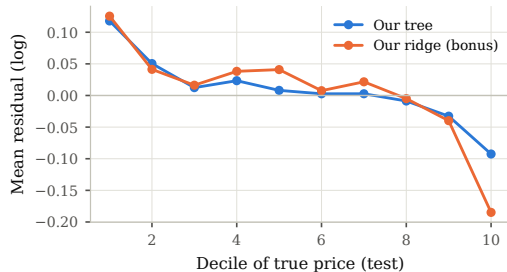
7. What the models use

- Tree importances (MDI): dist_center_km (0.59), land_area_sot (0.10), lng (0.08), area_m2 (0.04)
- Largest linear-SVM weights (standardised): + log_area, location=ag seher q., location=nizami m., total_floors; – dist_center_km, area_m2, location=bineqedi q., is_top_floor



8. Where the models fail, and limitations

- Luxury tail: the tree under-predicts the top decile (-0.093 log)
- Sparse districts: MAE 0.365 vs. 0.131 in dense ones
- Tier errors are near-ties at the threshold
- Limitations: asking $\neq$ sale price; one snapshot (no time split); fixed cleaning rules; RFF depends on the seed; our tree is slower than Cython

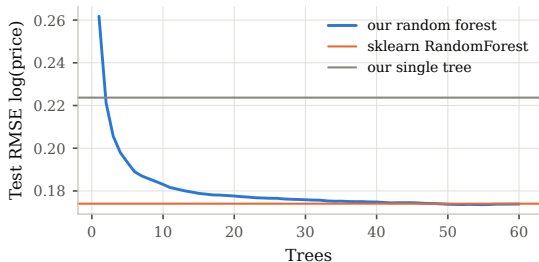


9. Bonus, contributions and next steps

- Ridge (normal equations): RMSE 0.231
- Our random forest: 0.174 (sklearn 0.174)
- Our gradient boosting: 0.200 (sklearn 0.201)

- **Who did what:** `contribution_report.md`
- Next: time-based split, in-fold target encoding, quantile loss for the tail

(a) Bagging + feature subsampling



(b) Least-squares boosting

